2017 Ph H2 Q11

Section: Particles and Waves

Topic: Wave-Particle Duality

Light is incident on a metal surface.
The maximum kinetic energy of the emitted photoelectrons is measured at different frequencies.

(a)

State what is meant by the threshold frequency.

✅ The **threshold frequency** is the **minimum frequency of incident radiation** required to **eject electrons** from the surface of a given metal.

> Below this frequency, **no photoelectrons are emitted**, regardless of intensity.

(b)

The graph below shows the relationship between the maximum kinetic energy of emitted photoelectrons and the frequency of incident light.

(Assume the graph is a straight line with a positive slope, intercepting the frequency axis at f_0.)

(i)

Use the graph to determine the work function of the metal in joules.

The work function ϕ is found from:

$$\phi = hf_0$$

From the graph (e.g. if $f_0 = 4.0 \times 10^{14}\ \mathrm{Hz}$):

$$\phi = 6.63 \times 10^{-34} \times 4.0 \times 10^{14} = 2.65 \times 10^{-19}\ \mathrm{J}$$

✅ *Revision tip:*

- The **x-intercept** of the graph gives the **threshold frequency**
- Use Planck's constant: $h = 6.63 \times 10^{-34}\ \mathrm{Js}$

(ii)

Use the graph to determine the value of Planck's constant.

Planck's constant is the **gradient** of the graph of E_k vs f:

$$h = \frac{\Delta E_k}{\Delta f}$$

Example from the graph:
If kinetic energy rises from 1.0×10^{-19} J to 4.0×10^{-19} J as frequency rises from 5.5×10^{14} Hz to 10.0×10^{14} Hz:

$$h = \frac{4.0 \times 10^{-19} - 1.0 \times 10^{-19}}{10.0 \times 10^{14} - 5.5 \times 10^{14}} = \frac{3.0 \times 10^{-19}}{4.5 \times 10^{14}} = 6.67 \times 10^{-34}\ \mathrm{Js}$$

(Which is very close to the known value.)

🧠 Revision Tips:

- The **photoelectric equation** is:
$E_k = hf - \phi$
- Graph of E_k vs f has:
  - **slope** = h
  - **x-intercept** = f_0
- Work function is the **minimum energy** needed to free an electron